Comparative Evaluation of Statistical Classifiers for Credit Card Default Prediction: A Threshold Analysis

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***Abstract*—Credit card default prediction is a binary classification problem in which overall accuracy can obscure a model's ability to identify the less frequent default class. This study compares Logistic Regression (LR), Linear Discriminant Analysis (LDA), and Quadratic Discriminant Analysis (QDA) on the UCI Default of Credit Card Clients dataset, comprising 30,000 observations and 23 predictors. The target contains 22.12% default cases. The models are evaluated on a stratified 80:20 train–test split using accuracy, precision, recall, F1-score, and receiver operating characteristic area under the curve (ROC-AUC). A separate experiment selects a QDA classification threshold using F1-score on a validation subset, after which QDA is refitted on the full training partition and evaluated on the untouched test set. At a threshold of 0.50, LDA has the highest accuracy (0.8087), while QDA has higher recall (0.6209) and F1-score (0.4953). Using the validation-selected QDA threshold of 0.75 increases test precision from 0.4120 to 0.4687 and F1-score from 0.4953 to 0.5071, while recall decreases from 0.6209 to 0.5524. ROC-AUC remains 0.7256 because the score ranking is unchanged. The results illustrate that model interpretation depends on the evaluation criterion and that threshold selection changes the operating point. The findings are limited to one dataset and one split and do not establish statistical significance or financial utility.**

***Index Terms*—credit risk, credit card default, Logistic Regression, Linear Discriminant Analysis, Quadratic Discriminant Analysis, imbalanced classification, decision threshold**

I. Introduction

Credit scoring uses customer information to estimate credit risk and support decisions about lending and account management. Statistical classification has long been used in consumer credit scoring, although model performance depends on the data, assumptions, and criteria used to evaluate predictions [1]–[3]. For credit default prediction, correctly identifying customers who will default may matter differently from correctly identifying customers who will not. Consequently, accuracy alone may not adequately describe performance when the target classes are imbalanced.

This study uses the UCI Default of Credit Card Clients dataset, originally associated with research on the probability of default among Taiwanese credit card clients [5], [6]. The dataset contains 30,000 observations and 23 predictors, with 6,636 default cases (22.12%). Three statistical classifiers are evaluated: Logistic Regression (LR), Linear Discriminant Analysis (LDA), and Quadratic Discriminant Analysis (QDA). The first comparison uses a conventional threshold of 0.50. A separate experiment selects a QDA threshold using validation F1-score and evaluates that threshold on a held-out test set.

The study has three objectives: (1) compare LR, LDA, and QDA at a fixed threshold using multiple metrics; (2) measure how a validation-selected QDA threshold changes test-set precision, recall, accuracy, and F1-score; and (3) examine how conclusions change when models are judged by different metrics. The contribution is an empirical, protocol-specific comparison rather than a new algorithm or a claim of universal model superiority.

The research questions are:

- **RQ1:** At threshold 0.50, how do LR, LDA, and QDA differ in accuracy, precision, recall, F1-score, and ROC-AUC on the selected dataset?
- **RQ2:** How does selecting QDA's threshold by validation F1-score affect its performance on the untouched test set relative to threshold 0.50?
- **RQ3:** How does the choice of evaluation metric change the interpretation of classifier performance under the observed class imbalance?

II. Related Work

A. Statistical credit scoring

Hand and Henley reviewed statistical classification methods used in consumer credit scoring and discussed methodological issues particular to this domain [1]. Thomas surveyed credit and behavioural scoring, including the distinction between estimating default risk and supporting economically meaningful decisions [2]. Benchmark studies have compared classifiers across credit datasets. Baesens *et al.* found that simple statistical methods such as LR and LDA remained competitive in their benchmark, while Lessmann *et al.* revisited credit-scoring benchmarks with newer methods and evaluation procedures [3], [4]. These studies support comparative evaluation but do not imply that a single classifier will be best across all datasets or objectives.

B. Default prediction using the UCI dataset

Yeh and Lien compared data-mining methods on the Taiwanese credit card data and emphasized the importance of estimating default probabilities, rather than considering only a final binary label [5]. The dataset is publicly available

through the UCI Machine Learning Repository [6]. Later work has evaluated statistical and machine-learning approaches on the same data. Bhandary and Ghosh compared LDA and LR with several more complex methods and reported differences across accuracy, F1-score, G-mean, and AUC under their experimental protocol [7]. Zhou examined cost-sensitive threshold adjustment for credit card default prediction using gradient boosting [8]. These studies establish that neither model choice nor decision threshold should be discussed without reference to the evaluation design. Their numerical results are not directly comparable with the present results because experimental splits, preprocessing, model families, and selection procedures differ.

C. Imbalance and evaluation metrics

Imbalanced learning has been widely studied, including in credit scoring [9], [10]. When non-default cases are the majority, a model can achieve a seemingly strong accuracy while missing many defaults. Precision measures the fraction of predicted defaults that are actual defaults, whereas recall measures the fraction of actual defaults identified. The F1-score is the harmonic mean of precision and recall. ROC analysis evaluates score-based discrimination across thresholds [11]; precision–recall analysis can add useful information when the positive class is relatively rare [12]. AUC also has known interpretive limitations, including a critique that it may not align coherently with a common misclassification-cost distribution across classifiers [13]. Accordingly, this study reports several metrics and interprets them jointly.

D. Threshold selection and evaluation protocol

A probability score and a binary decision are distinct: a threshold converts scores into predicted classes. Cost-sensitive decision theory shows why an appropriate cutoff depends on the relative consequences of different errors rather than being universally fixed at 0.50 [14]. F1-based thresholding has also been studied, with important qualifications concerning probability calibration and the objective itself [15]. Selection decisions made using a finite sample can overfit the selection criterion, which motivates keeping model or threshold selection separate from final evaluation [16]. Profit-based cutoffs have been studied in consumer credit scoring, but the present work does not estimate monetary utility [17].

III. Research Gap and Contribution

Existing literature has extensively examined credit scoring, comparisons among classifiers, imbalanced classification, and threshold selection. Threshold tuning on the UCI credit-card dataset has also been studied in a cost-sensitive boosting setting [8]; therefore, threshold adjustment itself is not claimed as novel. Within the sources reviewed for this paper, we did not identify a directly comparable study that isolates validation-F1 selection for regularized QDA and reports its held-out precision–recall trade-off alongside LR and LDA evaluated at the conventional threshold. This is a scoped gap in the comparison assembled here, not proof that no such study exists.

The study's contribution is threefold: a reproducible comparison of three statistical classifiers under one stratified split; a validation-based QDA threshold experiment with a separate test evaluation; and an interpretation of how accuracy, F1-score, recall, and ROC-AUC lead to different assessments. It does not propose a new classifier, establish a universal ranking, or measure lending profitability.

TABLE I
Predictor groups in the dataset.

Group	Examples of information represented
Credit information	Assigned credit limit.
Demographics	Age and coded demographic categories.
Repayment history	Monthly repayment-status indicators.
Bill statements	Historical monthly bill amounts.
Payments	Historical monthly payment amounts.

IV. Dataset and Exploratory Analysis

The UCI dataset contains 30,000 customer records and 23 predictors. The target variable, denoted Y in the downloaded data, indicates whether the customer defaulted in the following month. There are 23,364 non-default observations (77.88%) and 6,636 default observations (22.12%). The predictors cover credit limit, demographic characteristics, repayment status, bill statement amounts, and payment amounts [5], [6]. The source data describe account and repayment behaviour over a historical period in 2005. These variables provide different views of a customer's available credit and recent payment behaviour, but this study does not perform a separate causal analysis or claim that any single variable causes default.

Table I summarizes the predictor groups at a conceptual level. The table is descriptive; the models use the predictor columns supplied by the dataset rather than a manually selected subset.

The data represent a specific Taiwanese customer population and historical period; conclusions should not be assumed to generalize to other institutions, countries, or periods.

Figure 1 shows the class distribution. The majority-class baseline accuracy is approximately 0.7788 if every case is predicted as non-default. This baseline provides context for accuracy but does not, by itself, capture the utility of identifying defaults. The provided dataset audit code checks missing values, duplicate rows, repeated predictor patterns, and feature ranges. The UCI repository reports no missing values; median imputation is retained in the model pipelines as a preprocessing safeguard. Standardization and imputation are fitted inside each training pipeline rather than on the full dataset.

V. Methodology

A. Data split and preprocessing

The data are divided into a stratified 80:20 training–test split with `random_state=42`. This yields 24,000 training ob-

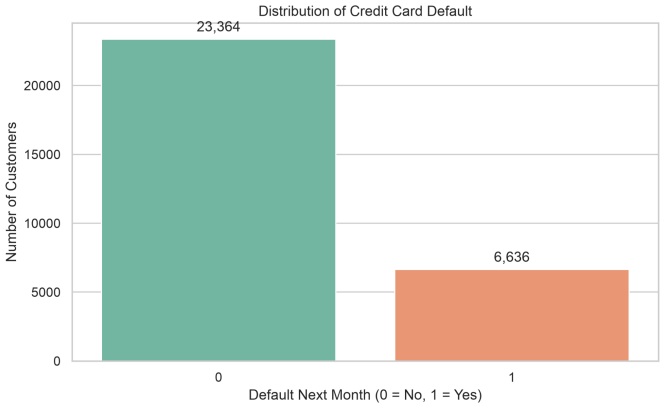


Fig. 1. Distribution of the target class in the UCI dataset.

servations and 6,000 test observations. Stratification preserves the approximate target-class proportion across partitions. Each classifier is implemented in a scikit-learn pipeline with median imputation and standard scaling followed by the classifier. Since the UCI data are reported to contain no missing values, median imputation is not expected to change the observed data; it remains part of the pipeline. Preprocessing is fitted using training data only.

B. Logistic Regression

Logistic Regression models the probability of default as

$$P(Y = 1 \mid \mathbf{x}) = \sigma(\beta_0 + \boldsymbol{\beta}^T \mathbf{x}), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \qquad (1)$$

where $\mathbf{x}$ is the predictor vector, β_0 is the intercept, and $\boldsymbol{\beta}$ is the coefficient vector. The fitted probability is converted to a class label using the selected threshold. The implementation uses `max_iter=2000` and `random_state=42`.

C. Linear and Quadratic Discriminant Analysis

LDA and QDA use class-conditional distribution estimates to classify observations. Under a multivariate Gaussian formulation, a discriminant score for class k can be written, up to terms common to classes, as

$$\delta_k(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\Sigma}_k^{-1} \boldsymbol{\mu}_k - \frac{1}{2} \boldsymbol{\mu}_k^T \boldsymbol{\Sigma}_k^{-1} \boldsymbol{\mu}_k + \log \pi_k, \qquad (2)$$

where $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}_k$ are the class mean and covariance estimates, and π_k is the class prior. In LDA, the covariance is shared across classes, so the quadratic terms in $\mathbf{x}$ cancel when comparing class scores and the boundary is linear. In QDA, class-specific covariance matrices remain, resulting in a quadratic boundary. QDA is therefore more flexible but estimates more parameters, which can make its predictions sensitive to covariance estimation. The implementation uses `reg_param=0.1` to regularize QDA's covariance estimates. The adequacy of the Gaussian assumptions is not separately tested.

Table II summarizes the main distinction relevant to this experiment.

TABLE II
CONCEPTUAL COMPARISON OF THE EVALUATED CLASSIFIERS.

Model	Main assumption / form	Interpretive point
LR	Linear log-odds in predictors	Probability-based baseline; decision threshold is separate.
LDA	Shared class covariance	Linear discriminant boundary under Gaussian assumptions.
QDA	Class-specific covariance	More flexible boundary; estimates more covariance parameters.

D. Threshold selection for QDA

The initial comparison evaluates all three classifiers at threshold 0.50. For the additional QDA experiment, the 24,000-observation training partition is divided into a subtraining set of 19,200 observations and a validation set of 4,800 observations, using a stratified split with `random_state=42`. QDA is fitted on the subtraining partition. Validation probabilities are classified using thresholds from 0.20 to 0.80 in increments of 0.05. The threshold with the highest validation F1-score is selected; the recorded selection is 0.75, with validation F1-score 0.5137.

After selection, QDA is refitted on the full 24,000-observation training set. Its test probabilities are evaluated once using thresholds 0.50 and 0.75. The test set is not used to select the threshold. The threshold search is performed only for QDA, so the comparison between QDA at 0.75 and LR/LDA at 0.50 combines a model difference with a threshold difference.

The protocol separates the experiments into two stages. Stage one provides a like-for-like fixed-threshold comparison on the same test observations. Stage two investigates threshold selection for QDA alone. In the second stage, the QDA model used to evaluate the validation grid is not the final test model: after the threshold is selected, QDA is fitted again using all training observations. This refitting uses training data only and does not expose the test labels to the selection procedure. The protocol is reproducible given the dataset, split seed, preprocessing pipeline, classifier parameters, and threshold grid, although a single split does not quantify variation across possible partitions.

Table III summarizes the stages and the role of each data partition.

E. Evaluation metrics

Accuracy is the fraction of all observations classified correctly. Precision and recall for the positive class (default, $Y = 1$) are

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}. \qquad (3)$$

The F1-score is

$$F_1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \qquad (4)$$

TABLE III
EXPERIMENTAL PROTOCOL AND PARTITION USAGE.

Stage	Data used	Purpose
Baseline comparison	24,000 train; 6,000 test	Fit LR, LDA, QDA; evaluate at 0.50.
Threshold selection	19,200 subtrain; 4,800 validation	Select QDA threshold by F1 over 13 candidates.
Final QDA evaluation	Refit on 24,000 train; test on 6,000	Evaluate QDA at 0.50 and 0.75 without test-based selection.

TABLE IV
RECORDED TEST-SET PERFORMANCE.

Model / threshold	Acc.	Prec.	Rec.	F1	AUC
LR / 0.50	0.8077	0.6868	0.2396	0.3553	0.7076
LDA / 0.50	0.8087	0.6801	0.2547	0.3706	0.7017
QDA / 0.50	0.7202	0.4120	0.6209	0.4953	0.7256
QDA / 0.75	0.7625	0.4687	0.5524	0.5071	0.7256

ROC-AUC measures ranking discrimination from the continuous predicted probabilities, across possible thresholds. Unlike accuracy, precision, recall, and F1 at a fixed threshold, ROC-AUC does not change merely because a different threshold is applied to the same probabilities. No monetary cost function or statistical significance test is included in the current experiments.

VI. RESULTS

A. Model comparison at threshold 0.50

Table IV summarizes the recorded test-set results. LDA has the highest accuracy (0.8087), narrowly ahead of LR (0.8077). LR has precision 0.6868, and LDA has precision 0.6801. Both models have low recall (0.2396 and 0.2547), indicating that many actual defaults are not identified at the conventional threshold. QDA at 0.50 has lower accuracy (0.7202) and precision (0.4120), but substantially higher recall (0.6209) and F1-score (0.4953). Its ROC-AUC of 0.7256 is also the highest among the three models.

The majority-class baseline accuracy is approximately 0.7788. LR and LDA exceed this baseline by about three percentage points, while both reported QDA operating points fall below it. This does not make QDA uninformative: its higher recall and F1-score indicate a different trade-off in identifying default cases. However, accuracy and the minority-class metrics answer different questions, and the preferred operating point depends on the decision objective.

B. QDA threshold analysis

The validation analysis selected threshold 0.75 by F1-score. On the held-out test set, increasing the threshold from 0.50 to 0.75 raises precision from 0.4120 to 0.4687 and accuracy from 0.7202 to 0.7625. Recall falls from 0.6209 to 0.5524, while F1-score increases from 0.4953 to 0.5071. The absolute F1 difference is 0.0118 and should be described as a modest

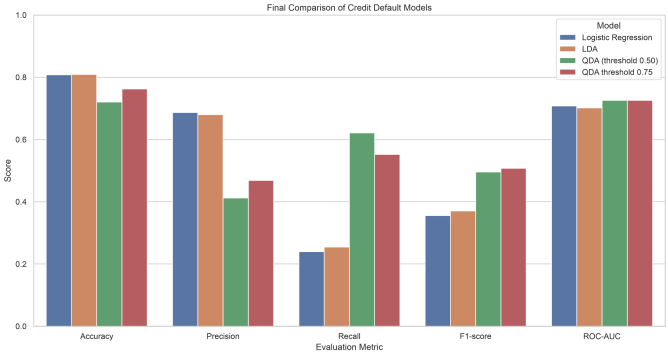


Fig. 2. Comparison of the recorded test metrics for LR, LDA, and QDA at two thresholds.

observed change; no uncertainty interval or significance test was performed.

ROC-AUC remains 0.7256 at both thresholds because it is calculated from the same continuous test probabilities. The threshold changes which observations are labelled positive, but it does not alter the ranking of those probabilities. Figure 3 shows the validation precision, recall, and F1-score across candidate thresholds.

VII. DISCUSSION

The results answer RQ1 by showing that no single classifier leads on every metric. At threshold 0.50, LDA provides the highest accuracy, whereas QDA yields higher recall and F1-score. The results answer RQ2 by showing that the validation-selected QDA threshold of 0.75 changes the test operating point: precision and F1-score increase, while recall decreases. For RQ3, the model ordering changes with the metric: accuracy favours LDA, F1-score favours QDA at 0.75, and ROC-AUC favours QDA's underlying score ranking. This is why the paper reports several complementary measures instead of treating accuracy as a sufficient summary.

The threshold change reduces the number of observations predicted as default, which is consistent with the rise in precision and fall in recall. This pattern follows from the threshold rule: a higher cutoff requires a larger estimated probability before assigning the positive label. Some cases previously labelled as defaults are consequently reclassified as non-default. This can reduce false-positive predictions while increasing false negatives. The present metrics summarize that trade-off, but without an explicit cost matrix they cannot determine whether the change is desirable for a lender or customer.

The F1-score is useful here because it combines precision and recall, but it is not a universal decision objective. It assigns equal formal importance to precision and recall through their harmonic mean and does not include the prevalence-independent monetary cost of either error. Likewise, ROC-AUC summarizes ranking across thresholds but does not specify the operating point to use. Because the study does not evaluate probability calibration, a predicted probability should not be interpreted as a verified real-world default frequency for

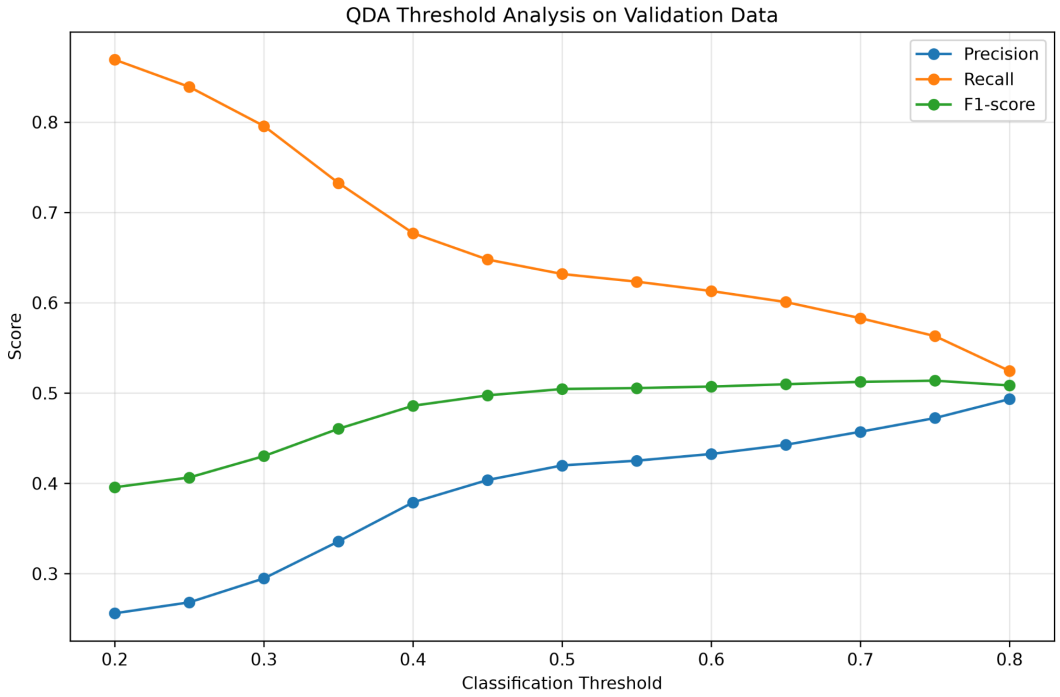


Fig. 3. QDA precision, recall, and F1-score across thresholds on validation data.

a customer. These distinctions are important when translating a classroom classification experiment into a financial decision setting. The study does not calculate the financial cost of either type of error, so it cannot establish whether the higher threshold is economically preferable.

The results should also be interpreted in relation to earlier studies without treating their reported scores as a leaderboard. Bhandary and Ghosh evaluate a broader set of statistical and machine-learning methods using a different experimental design, including cross-validation and variable shortlisting [7]. Zhou considers a cost-sensitive threshold in a boosting-based approach [8]. These studies support the relevance of comparing models and decision thresholds, but their scores cannot be directly compared with the single hold-out results reported here. The present study is narrower: it focuses on LR, LDA, and QDA and reports a threshold selected on validation data, followed by one test evaluation. The distinction is important because model family, feature selection, resampling, split strategy, and threshold selection can all affect observed metrics.

A second interpretation concerns the majority-class baseline. An accuracy near 0.779 is obtained by predicting non-default for every customer, but that rule has zero recall for default. LR and LDA improve accuracy over this baseline while detecting only around one quarter of actual defaults at threshold 0.50. QDA detects more defaults but predicts the positive class more often, reducing overall accuracy. This illustrates a trade-off rather than an unconditional ranking. The appropriate operating point depends on the purpose of the system and the relative costs of false alarms and missed defaults, neither of which is quantified in this experiment. In operational credit decisions, the relative consequences of false negatives and false positives would need to be specified before choosing a decision threshold. Further, QDA's threshold was tuned while LR and LDA were not. The current comparison therefore does not establish that QDA is intrinsically better than LR or LDA under a matched threshold-selection protocol.

The reported threshold of 0.75 may motivate a future probability-calibration analysis, but it does not itself prove that QDA probabilities are poorly calibrated. Calibration was not measured in this study. Likewise, the experiments use one stratified split; no repeated-split, cross-validation, confidence-interval, or significance analysis was performed. The observed metric differences should therefore be interpreted descriptively.

VIII. Limitations and Future Work

Several limitations constrain the conclusions. First, the dataset represents a particular Taiwanese customer population and historical period, so external validity is limited. Second, only three statistical classifiers are compared. Third, the model comparison uses one train–test split, and the QDA threshold experiment does not apply an equivalent threshold search to LR and LDA. Fourth, threshold selection optimizes validation F1-score rather than a financial cost function. F1 does not encode the actual costs of missed defaults and false alarms. Fifth, the current analysis does not assess probability calibration, confidence intervals, or statistical significance.

Future work should apply the same validation-based threshold-selection protocol to LR and LDA to separate model effects from threshold effects. Repeated splits or cross-

validation could measure sensitivity to the particular partition. Calibration curves or Brier scores could test whether predicted probabilities align with observed default frequencies. A cost-sensitive evaluation could be considered if defensible error costs were available. These are proposed extensions and were not performed in the present study.

IX. Conclusion

This study compared LR, LDA, and QDA for credit card default prediction using the UCI Default of Credit Card Clients dataset. At a threshold of 0.50, LDA achieved the highest accuracy, while QDA produced substantially higher recall and F1-score. Selecting QDA's threshold on validation data and applying it to the untouched test set increased precision and F1-score but reduced recall; ROC-AUC remained unchanged. The results demonstrate that classifier evaluation depends on the metric and the selected decision threshold. Because the evidence comes from one dataset and one split, with threshold tuning performed only for QDA, the findings are descriptive rather than universal. A fairer threshold comparison across all three models and an assessment of uncertainty and calibration are natural next steps.

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